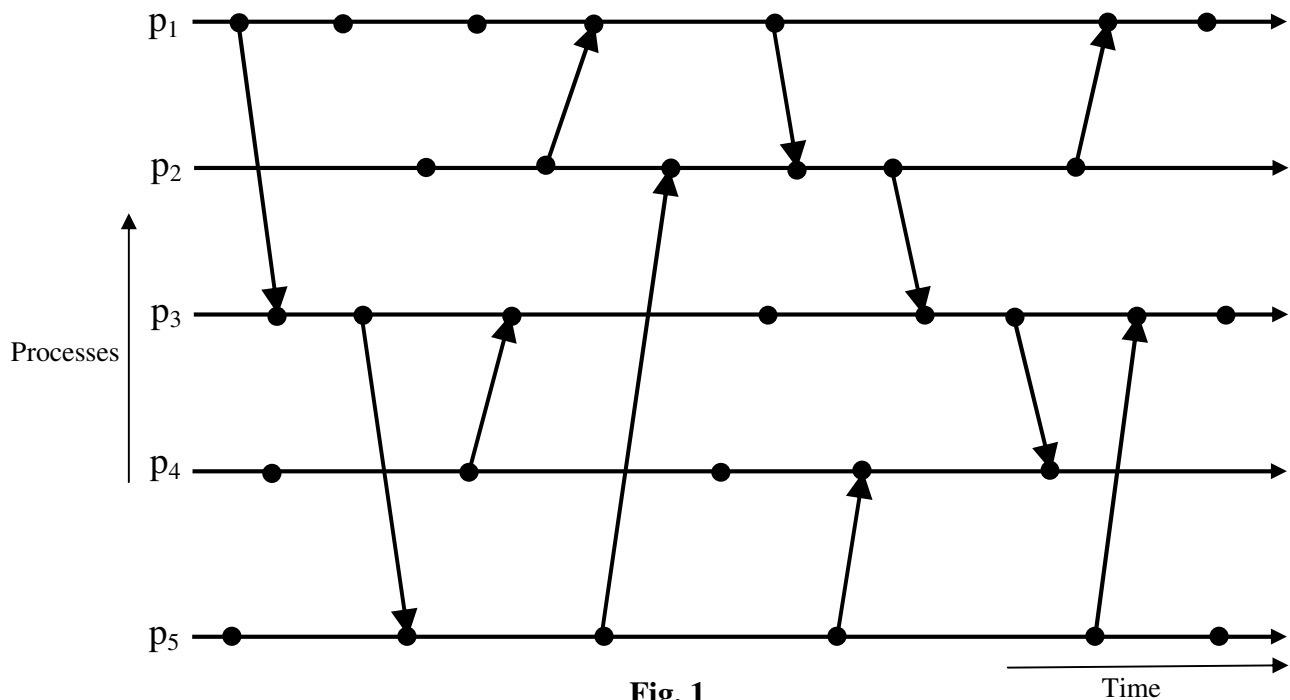


No. of Pages = 5
No. of Questions = 5

Assumptions made if any, should be stated clearly at the beginning of your answer.

- Q.1 (a) Identify the scalar clock values of the events and the timestamps of the messages exchanged among the 5 processes shown in Fig. 1. Initially, the scalar clocks of all the 5 processes are 0. Use **R1** and **R2** for updating the scalar clocks. Assume $d = 1$. [5]



Time

- Q.1 (b) Compute the vector timestamps of the events occurring at the 5 processes and the vector timestamps of the messages exchanged among them as shown in the space-time diagram of Fig. 1. Use **R1** and **R2** for updating the vector clocks. Initially, the vector clock of each of the 5 processes is $[0\ 0\ 0\ 0\ 0]$ and $d = 1$. [5]

- Q.2 (a) Consider the money transfer scenario shown in the timing diagram of Fig. 2. S1 and S2 are two sites of a distributed system which maintain bank accounts X and Y respectively. C_{12} is the communication channel from site S1 to site S2 and C_{21} is the communication channel from site S2 to site S1. The dashed vertical lines denote the different time instants of this timing diagram. Initially, at time instant t_0 , account X has \$500 and account Y has \$600. Between time instants t_0 and t_1 , S2 initiates a transfer of \$10 from account Y to account X. Between time instants t_3 and t_4 , S2 again initiates a transfer of \$30 from account Y to account X. S1 receives the first money transfer message after t_1 and the second money transfer message after t_4 . Between time instants t_1 and t_2 , S1 initiates a transfer of \$40 from account X to account Y. S2 receives this money transfer message after t_4 . Between time instants t_3 and t_4 , S1 initiates a transfer of \$90 from account X to account Y. S2 receives this money transfer message after t_5 . These 4 money transfer messages are shown by the 4 bold arrows in the diagram.

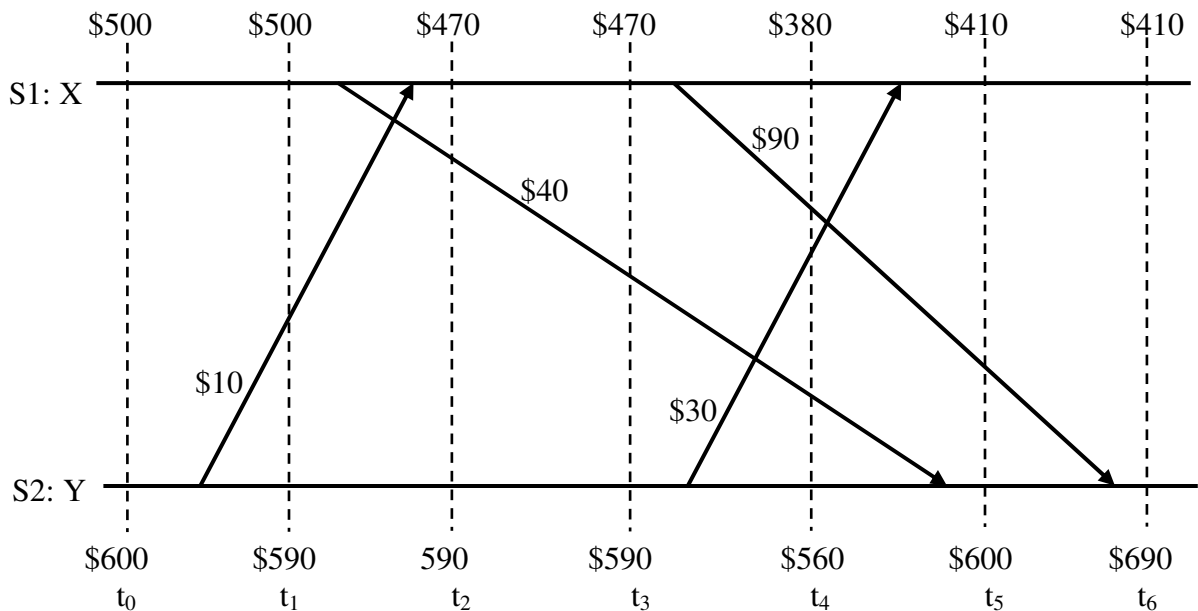


Fig. 2

- i) Determine the states of channels C_{12} and C_{21} at the different time instants for Fig. 2. [3.5]
- ii) Assume that the Lai-Yang algorithm is executed on this distributed system for recording the global snapshot. S2 initiates the snapshot recording algorithm between time instants t_2 and t_3 . [4]

Q.2 (b) For a distributed system, the synchronization delay is 0.01 seconds and the average critical section execution time is 5 seconds. What is the throughput of the system? [0.5]

Q.2 (c) Determine a BFS spanning tree by applying the synchronous single-initiator spanning tree algorithm using flooding on the graph shown in Fig. 3. Assume node A to be the root node. Clearly show the round numbers in which the QUERY messages are sent by the different nodes. [2]

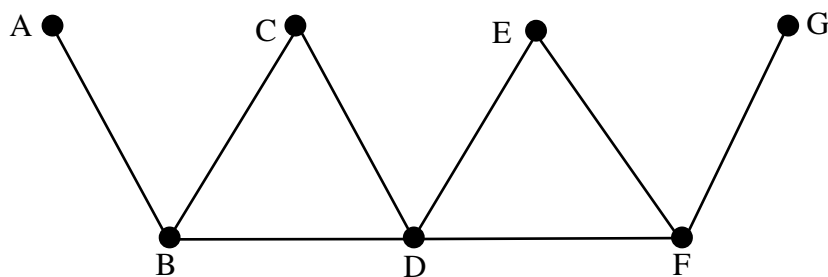


Fig. 3

- Q.3 (a) Fig. 4 shows a distributed system consisting of 4 sites - S_1 , S_2 , S_3 and S_4 . Assume that initially, the scalar clock values of events a (occurring at S_2) and b (occurring at S_4) are 2 and 1 respectively. Both S_2 and S_4 wish to enter the critical section and broadcast timestamped REQUEST messages. Execute Lamport's algorithm for implementing distributed mutual exclusion on this system. Clearly show the exchange of REPLY and RELEASE messages among the sites, the contents of request_queue of each of the sites corresponding to the send and receive events of the REQUEST and the RELEASE messages and the events when each of the sites enter and exit the critical section. [5]

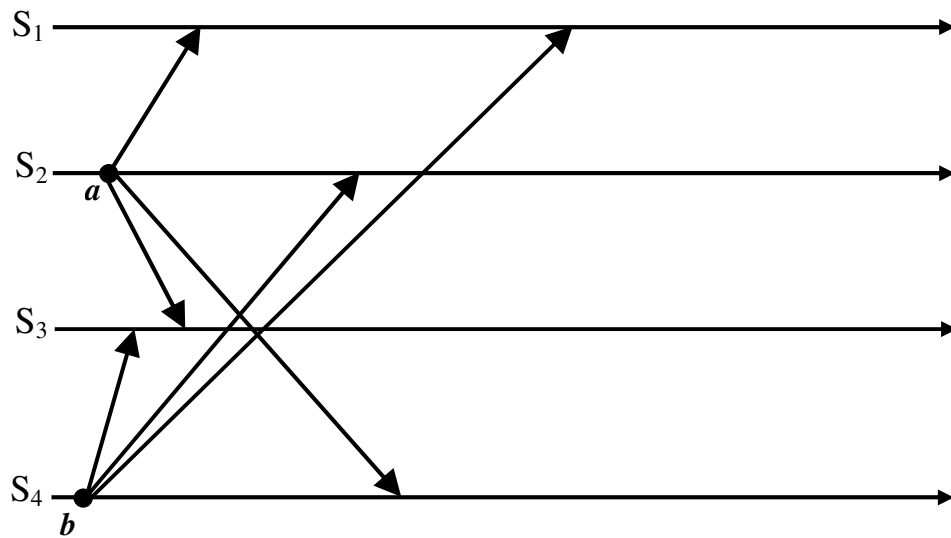


Fig. 4

- Q.3 (b) Consider the spanning tree shown in Fig. 5. Node N holds the privilege and is not currently executing the critical section. If node K wants to execute the critical section, using the knowledge of Raymond's tree-based algorithm, describe the sequence of events via which node K will receive the privilege. [5]

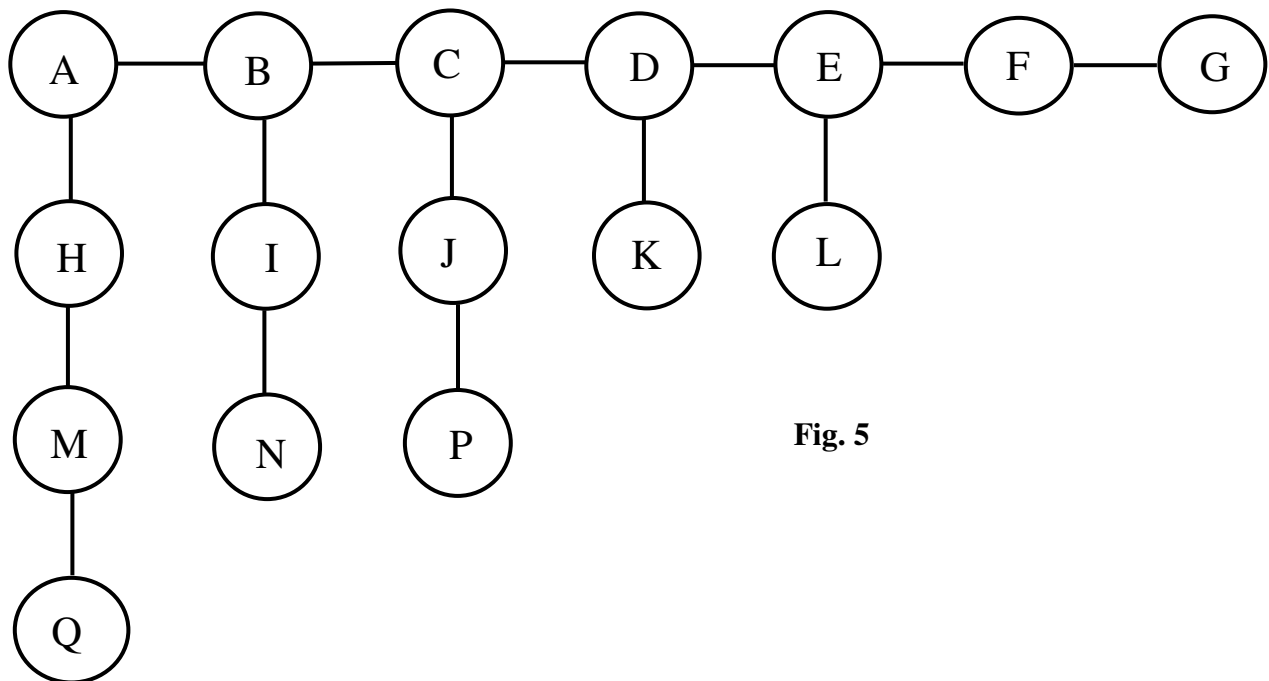


Fig. 5

Q.4 (a) Consider a distributed system consisting of 12 processes as depicted in Fig. 6.

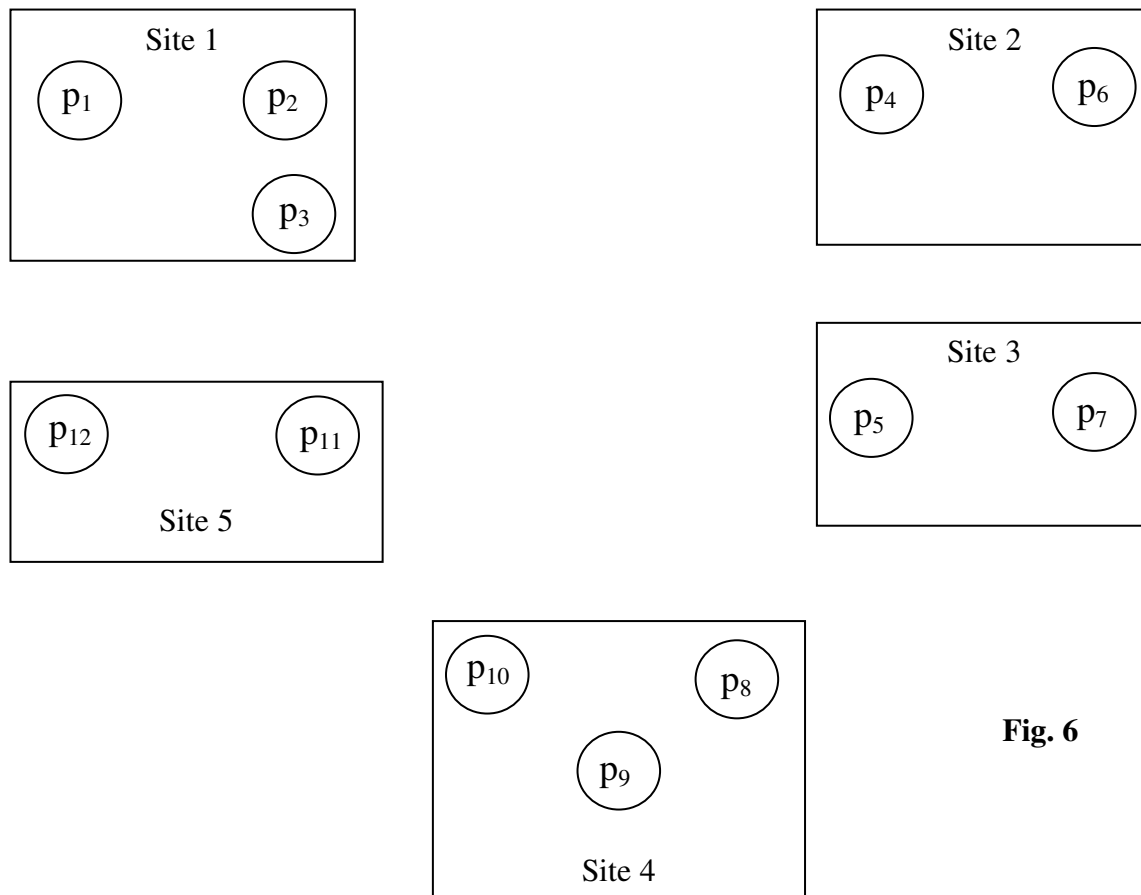


Fig. 6

The following dependencies exist among the processes:

- p₁ is blocked and is waiting for p₂ and p₁₂ to release some resources
- p₂ is blocked and is waiting for p₄ and p₅ to release some resources
- p₃ is blocked and is waiting for p₂ to release some resource
- p₄ is blocked and is waiting for p₆ to release some resource
- p₅ is blocked and is waiting for p₆ and p₇ to release some resources
- p₇ is blocked and is waiting for p₈ to release some resource
- p₈ is blocked and is waiting for p₉ to release some resource
- p₉ is blocked and is waiting for p₁₀ to release some resource
- p₁₀ is blocked and is waiting for p₁₁ to release some resource

- i) Draw the WFG for the above scenario. [1]
- ii) Identify the locally dependent processes. [1]
- iii) Assuming that the WFG determined in (i) is an AND model WFG, determine whether a deadlock exists in the system with proper reason. In case of a deadlock, identify the deadlocked processes. [2]
- iv) Assuming that the WFG determined in (i) is an OR model WFG, determine whether a deadlock exists in the system with proper reason. Justify your answer. [3]

- Q.4 (b) Suppose a synchronous system consists of 12 processes. Assume that the Byzantine agreement tree algorithm is executed on this system.
- i) What is the maximum number of Byzantine processes that can be tolerated so that the Byzantine agreement problem can be solved? [0.5]
 - ii) What will be the total number of rounds for the algorithm? [0.5]
 - iii) What are the number of messages exchanged in the 1st and the 3rd rounds? [1]
 - iv) A message will visit how many nodes in the 2nd round? [0.5]
 - v) Each message will be sent to how many nodes in the last round? [0.5]
- Q.5 (a) For a Chord ring of $m = 6$, there are 9 nodes – N3, N8, N15, N20, N29, N36, N48, N54 and N60 and 8 keys – K4, K8, K10, K25, K30, K32, K50 and K60.
- i) Show with the help of a diagram, the arrangement of the nodes along the Chord ring. [1]
 - ii) Identify on which node each of the above mentioned keys will be placed for this Chord ring. You should clearly mention the reason for placing each key on a particular node. [0.5 x 8 = 4]
 - iii) Suppose for this Chord ring, a lookup for K10 is initiated at N36. Describe the step-by-step procedure by which K10 can be located using the simple lookup algorithm. [3]
- Q.5 (b) Consider the list $L = \langle N1, N2, N3, N5, N8, N9, N10, N11, N14, N15, N16 \rangle$. Determine the outputs of the following operations. [0.5 x 4 = 2]
- i) head(L)
 - ii) tail(L)
 - iii) head(tail(tail(L)))
 - iv) tail(tail(tail(tail(L))))
